Multi-stage Evidence Retrieval and Claim Classification for Climate Fact-Checking

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Abstract

Climate claim verification requires retrieving relevant evidence from a large knowledge source and classifying each claim as SUPPORTS, REFUTES, NOT_ENOUGH_INFO, or DISPUTED. These two components are closely connected, since incomplete or inaccurate evidence retrieval can directly limit downstream classification performance.

We propose a multi-stage climate fact-checking pipeline that combines dual TF-IDF candidate retrieval, Cross-Encoder re-ranking, score fusion, adaptive evidence selection, and Transformer-based claim classification. The system first constructs a broad lexical candidate pool, re-ranks candidate passages using semantic claim-evidence relevance, and then selects the final evidence set using an adaptive strategy rather than a fixed top-k rule. The selected evidence passages are concatenated with the claim and passed to a Transformer-based classifier for four-way label prediction.

Our main finding is that training the classifier on retrieved rather than gold evidence improves classification accuracy from 0.4814 to 0.5649, by better matching the noisy retrieval setting at inference time. On the development set, the final system achieved an evidence F-score of 0.2383, classification accuracy of 0.5649, and harmonic mean of 0.3352.

1 Introduction

Climate change is a socially important domain in which misleading or unsupported claims can influence public understanding of scientific evidence. This project addresses automated climate claim verification, a task related to fact extraction and verification benchmarks such as FEVER (Thorne et al., 2018). Given a climate-related claim and a large evidence corpus, the system must retrieve relevant evidence passages and classify the claim as SUPPORTS, REFUTES, NOT_ENOUGH_INFO, or DISPUTED. The task therefore requires both evidence retrieval and claim classification.

The task is challenging because retrieval and classification are strongly interdependent. The corpus is too large for exhaustive neural scoring, so an efficient first-stage retriever is required. However, lexical retrieval may miss relevant evidence expressed with different wording, creating a retrieval bottleneck: if relevant evidence is absent from the candidate pool, later stages cannot recover it. Classification is further affected by noisy retrieved evidence and label imbalance, especially for ambiguous labels such as NOT_ENOUGH_INFO and DISPUTED.

We propose a multi-stage pipeline that combines lexical candidate retrieval, Cross-Encoder re-ranking, score fusion, adaptive evidence selection, and Transformer-based claim classification. The retrieval stage follows the common use of sparse term-weighting methods for efficient first-stage information retrieval (Manning et al., 2008), while the re-ranking stage follows BERT-style passage re-ranking methods that score query-passage pairs after initial retrieval (Devlin et al., 2019; Nogueira and Cho, 2019). This design reduces the search space before applying neural models to a smaller set of candidate passages.

For claim classification, the selected evidence passages are concatenated with the claim and passed to a Transformer-based classifier based on DeBERTa (He et al., 2021). A key design choice is to train the classifier using retrieved evidence rather than gold evidence, so that training better matches the noisy evidence available at inference time.

On the development set, the final system achieved an evidence F-score of 0.2383, classification accuracy of 0.5649, and harmonic mean of 0.3352. The results indicate that Cross-Encoder re-ranking improves evidence retrieval, and that training the classifier on retrieved evidence improves robustness to the noisy inference setting.

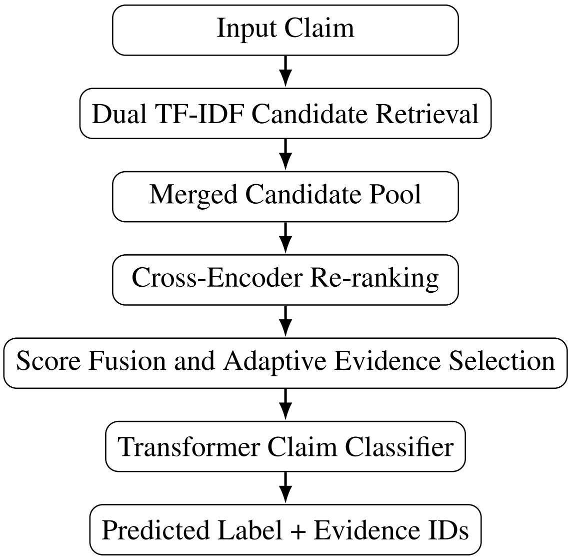

Figure 1: Overview of the proposed multi-stage climate claim verification pipeline.

2 Approach

2.1 System Overview

Our system is a multi-stage pipeline for automated climate claim verification, as illustrated in Figure 1. Given a claim, the pipeline proceeds through four stages: dual TF-IDF candidate retrieval, Cross-Encoder re-ranking, score fusion with adaptive evidence selection, and Transformer-based claim classification. This retrieve-then-verify structure follows the general evidence retrieval and claim verification setting used in prior fact verification work (Thorne et al., 2018; Diggelmann et al., 2020). The final output for each claim consists of a predicted label from {SUPPORTS, REFUTES, NOT_ENOUGH_INFO, DISPUTED} and a list of retrieved evidence passage IDs. Each stage is described in detail in the following subsections.

2.2 Candidate Evidence Retrieval

The candidate retrieval stage reduces the evidence corpus to a smaller set of potentially relevant passages. Since the corpus contains more than one million passages, it is not practical to apply a Cross-Encoder to every claim-evidence pair. We therefore use TF-IDF as an efficient first-stage retriever (Manning et al., 2008). We implement two complementary TF-IDF retrievers to improve candidate diversity. The first uses unigram features with English stopword removal to capture main content-word overlap. The second uses unigram and bigram features without stopword removal to preserve short phrase patterns and local word order. For each claim, we retrieve the top 250 passages from each retriever and merge them into a candidate pool of up to 400 unique passages. This pool is then passed to the Cross-Encoder re-ranking stage. Although this improves first-stage coverage, TF-IDF remains limited by surface word overlap and is therefore used only for candidate generation (Manning et al., 2008).

2.3 Cross-Encoder Evidence Re-ranking

After candidate retrieval, we use a Cross-Encoder to re-rank passages according to semantic relevance (Nogueira and Cho, 2019). The model takes a claim and an evidence passage as paired input and outputs a relevance score. Unlike TF-IDF, the Cross-Encoder jointly encodes the two sequences, allowing token-level interactions between the claim and evidence. It is applied only to the TF-IDF candidate pool rather than the full corpus, combining efficient lexical retrieval with stronger semantic matching.

We initialise the re-ranker from MiniLM-L-6 and fine-tune it using training claim-evidence pairs (Wang et al., 2020). This checkpoint was chosen as a practical trade-off between inference speed and ranking quality under limited GPU memory and runtime. Gold evidence passages are used as positive examples, while non-gold passages are used as negative examples. In the final setup, we sample four negative passages for each positive pair, producing a binary relevance training set.

2.4 Score Fusion and Adaptive Evidence Selection

The Cross-Encoder produces a semantic relevance score for each candidate claim-evidence pair. A simple approach would be to select the top-k passages directly according to this score. However, using the Cross-Encoder score alone may ignore useful lexical information from the first-stage retrievers. The Cross-Encoder assigns continuous relevance scores, and in practice several candidates often receive very similar scores, particularly near the selection boundary. In such cases, a small lexical signal can act as a tiebreaker: a candidate that also shares strong surface overlap with the claim is preferable to one that does not, even when their neural scores are nearly equal. Therefore, we combine the Cross-Encoder score with the lexical score from the retrieval stage.

Let c denote a claim and e denote a candidate evidence passage. We compute the final evidence score as:

$$s_{\text{final}}(c, e) = (1 - \alpha)s_{\text{ce}}(c, e) + \alpha s_{\text{lex}}(c, e)$$

where $s_{\text{ce}}(c, e)$ is the normalised Cross-Encoder score, $s_{\text{lex}}(c, e)$ is the normalised lexical retrieval score, and α controls the contribution of the lexical signal. In our final system, α is set to 0.03. This means that the Cross-Encoder remains the dominant ranking signal, while the lexical score is used as a weak stabilising component.

After score fusion, the system must decide how many evidence passages to return for each claim. A fixed top-k strategy assumes that all claims require the same number of evidence passages. This is not ideal for the fact-checking task. Some claims may be verified using one highly relevant passage, while other claims may require multiple passages, especially when the label is DISPUTED or when different evidence passages capture different parts of the claim. Selecting too many passages may reduce precision by adding noisy evidence, while selecting too few passages may reduce recall.

To address this issue, we use an adaptive evidence selection strategy based on the score gap from the top-ranked passage. Let $s_{\max}$ be the highest final score among the candidates for a claim. We select evidence passages whose final scores are close to this top score:

$$E_c = \{e_i \mid s_{\text{final}}(c, e_i) \geq s_{\max} - \delta\}$$

where δ is a gap threshold. In the final setting, we use $\delta = 0.03$. To ensure a valid and stable output, we also constrain the number of selected evidence passages using a minimum and maximum bound. Specifically, the system returns at least one evidence passage and at most six evidence passages for each claim. The selected evidence passages are then used as the evidence context for the claim classification module.

2.5 Claim Classification

After the final evidence set is selected, the system performs claim classification using a Transformer-based classifier. The goal of this module is to predict one of four labels: SUPPORTS, REFUTES, NOT_ENOUGH_INFO, or DISPUTED. Unlike a claim-only classifier, our classifier uses both the claim and the retrieved evidence passages, so that the prediction is grounded in the evidence selected by the retrieval module.

For each claim, the selected evidence passages are ordered according to their final retrieval scores and concatenated into a single evidence context. The classifier input is constructed by pairing the claim with this evidence context. This allows the model to jointly consider the claim and the retrieved evidence when making the final label prediction. If the combined input exceeds the maximum sequence length, the evidence context is truncated to fit the model input limit.

We use a DeBERTa-based Transformer classifier and fine-tune it on the provided training claims (He et al., 2021). In our final system, the classifier is initialised from a pretrained natural language inference checkpoint and adapted to the four-way claim classification task. The final classification layer predicts one of the four project labels, and the model is trained using square-root class-weighted cross-entropy loss.

A key design choice is that the classifier is trained using retrieved evidence rather than only gold evidence. During inference, gold evidence is not available, and the classifier must rely on the evidence selected by the retrieval module. Training with retrieved evidence therefore makes the training setting closer to the actual inference setting. However, this also means that classification performance depends on retrieval quality: if the selected evidence is incomplete or noisy, the classifier may receive insufficient information for the correct label decision.

The training labels are imbalanced, with SUPPORTS and NOT_ENOUGH_INFO appearing more frequently than REFUTES and DISPUTED. To reduce the tendency of the classifier to over-predict majority classes without over-correcting the class distribution, we use square-root class weights in the cross-entropy loss. Using the inverse class frequency directly as weights would amplify the weight of the DISPUTED class too aggressively given its small count, risking over-correction. Taking the square root provides a more conservative upweighting that encourages the model to attend to minority classes while still learning from the full training distribution (He and Garcia, 2009; Cui et al., 2019).

2.6 Baselines

We compare the final system with several baselines and ablation variants to isolate the contribution of each major component. The dummy majority baseline predicts the majority label for all claims and returns a fixed evidence passage, providing a lower bound for the evaluation pipeline.

For retrieval, we evaluate TF-IDF and BM25 lexical baselines to measure the effectiveness of surface word matching without neural re-ranking (Manning et al., 2008; Robertson and Zaragoza, 2009). We also compare a single TF-IDF candidate retriever with our dual TF-IDF retriever to assess whether combining complementary lexical views improves the candidate pool before Cross-Encoder re-ranking.

For evidence selection, we compare fixed top-k selection with our score fusion and adaptive selection strategy. This tests whether selecting a variable number of evidence passages based on score gaps provides a better precision-recall trade-off than returning the same number of passages for every claim.

For claim classification, we compare classifiers trained with gold evidence and retrieved evidence. This diagnostic comparison examines whether training on automatically retrieved passages better matches the test-time setting, where gold evidence is unavailable.

3 Experiments

3.1 Evaluation Method

We evaluate our system on the development set using the official evaluation script. The evaluation reports three metrics: evidence retrieval F-score, claim classification accuracy, and the harmonic mean of the two.

Evidence retrieval F-score measures how well the retrieved evidence IDs match the gold evidence IDs for each claim. For each claim, precision and recall are computed by comparing the predicted evidence set with the gold evidence set, and the F-score is averaged across all development claims.

Claim classification accuracy measures the proportion of claims whose predicted label matches the gold label. This metric evaluates the classification component independently of whether the retrieved evidence is correct.

The harmonic mean combines evidence retrieval F-score and claim classification accuracy:

$$H = \frac{2FA}{F + A}$$

where F is the evidence retrieval F-score and A is the claim classification accuracy. We use this metric to assess whether the system balances retrieval quality and classification performance rather than optimising only one component.

3.2 Experimental Details

We use the provided training, development, and test splits. The training set contains 1,228 labelled claims, the development set contains 154 labelled claims, the test set contains 153 unlabelled claims, and the evidence corpus contains 1,208,827 passages. The training set is used for model training, while the development set is used for hyperparameter selection and model comparison. The unlabelled test set is used only for final prediction generation and is not manually inspected.

We first tune a basic two-stage retrieval pipeline consisting of TF-IDF candidate retrieval followed by Cross-Encoder re-ranking with fixed top-k evidence selection. In this stage, we vary the number of retrieved candidates, the number of final evidence passages, the number of random negative examples per positive training pair, the number of Cross-Encoder fine-tuning epochs, and the Cross-Encoder learning rate. Specifically, we test $K_{\text{candidates}} \in \{300, 400\}$, fixed top-$k$ evidence selection with $k \in \{3, 4, 5\}$, negative examples per positive pair in {2,3,4,5}, fine-tuning epochs in {1,3,5}, and learning rates in $\{1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}\}$. The strongest initial setting, selected based on development evidence F-score, uses 300 retrieved candidates, fixed top-5 evidence selection, four random negatives per positive claim-evidence pair, three fine-tuning epochs, and a learning rate of 1×10^{-5}. The re-ranker is initialised from `cross-encoder/ms-marco-MiniLM-L-6-v2`.

After this initial search, we compare a single TF-IDF candidate retriever with the dual TF-IDF candidate retriever used in the final system. The single-retriever variant uses one TF-IDF model to retrieve the top 300 candidates for each claim. The dual-retriever variant retrieves the top 250 candidates from a unigram TF-IDF retriever with stopword removal and the top 250 candidates from a unigram-bigram TF-IDF retriever without stopword removal, then merges them into at most 400 unique candidates. To make this comparison fair, both variants use the same fine-tuned Cross-Encoder and fixed top-5 evidence selection. The dual-retriever setting achieves a higher development evidence F-score and is therefore used as the candidate retrieval method in the final system.

For score fusion and evidence selection, we search over several fusion and selection settings on the development set. To improve evidence retrieval, we test different selection strategies for converting the re-ranked candidate list into the final evidence set. We compare fixed top-k selection with adaptive evidence selection. For fusion, we search α values in $\{0.00, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08\}$. For fixed top-$k$ selection, we test $k \in \{2, 3, 4, 5, 6\}$. For adaptive selection, we test score-gap thresholds in $\{0.03, 0.05, 0.08, 0.10, 0.12, 0.15, 0.20\}$, absolute score thresholds in $\{0.55, 0.60, 0.65, 0.70, 0.75, 0.80\}$, minimum evidence counts in $\{1,2,3\}$, and maximum evidence counts in $\{3,4,5,6\}$. The final system uses gap-based adaptive selection with $\alpha = 0.03$, $\delta = 0.03$, minimum evidence count 1, and maximum evidence count 6.

For claim classification, we fine-tune `cross-encoder/nli-deberta-v3-small` as a four-way Transformer classifier (He et al., 2021; Williams et al., 2018). The classifier input consists of the claim and the selected evidence context. The model is trained for 5 epochs with learning rate 2×10^{-5}, batch size 8, maximum sequence length 512, warmup ratio 0.1, and square-root class-weighted cross-entropy loss. We select the best checkpoint based on development accuracy using retrieved evidence.

We also compare two evidence sources for classification: gold evidence and retrieved evidence. The gold-evidence setting is used as a diagnostic comparison to study the effect of evidence source, while the final system uses retrieved evidence because this matches the actual inference setting where gold evidence is unavailable.

4 Results and Analysis

4.1 Main Results

Table 1 presents the evidence retrieval F-score, claim classification accuracy, and harmonic mean for all baselines and system variants evaluated on the development set. Rows above the second horizontal rule use majority-label classification (SUPPORTS) and are included to isolate retrieval performance. The oracle row applies the classifier trained on gold evidence to the same retrieved passages as the final system, isolating the effect of the training evidence source. The retrieval F-score of the final system (0.2383) is slightly lower than the score fusion search result (0.2438) because the Cross-Encoder is retrained independently for the final run; the small difference arises from non-deterministic negative sampling due to Python `set()` iteration order across different runtime sessions.

System	F	A	H
Dummy majority baseline	0.0000	0.4416	0.0000
TF-IDF top-1	0.0978	0.4416	0.1602
TF-IDF top-3	0.0959	0.4416	0.1576
TF-IDF top-5	0.0883	0.4416	0.1471
TF-IDF top-10	0.0724	0.4416	0.1244
TF-IDF top-100 + BM25 rerank	0.0704	0.4416	0.1214
BM25 top-5	0.1053	0.4416	0.1701
Single TF-IDF + Cross-Encoder	0.2037	0.4416	0.2788
Dual TF-IDF + Cross-Encoder	0.2080	0.4416	0.2829
+ score fusion + adaptive sel.	**0.2438**	0.4416	0.3141
Final retrieval + majority label	0.2383	0.4416	0.3096
Oracle (retrieved evid. + gold clf.)	0.2383	0.4814	0.3188
Final system (retrieved evid. + retrieved clf.)	0.2383	**0.5649**	**0.3352**

Table 1: Development set results. F = evidence F-score, A = classification accuracy, H = harmonic mean.

4.2 Lexical Retrieval Baselines

The dummy majority baseline obtained zero evidence F-score, confirming that predicting a fixed evidence passage without any retrieval is uninformative, and establishing the majority-label classification accuracy of 0.4416 as a lower bound for the classification component.

TF-IDF retrieval substantially improved evidence F-score over the dummy baseline, reaching 0.0978 at top-1. However, F-score decreased monotonically as top-k increased from 1 to 10, falling from 0.0978 to 0.0724. This pattern indicates that retrieving more passages degraded precision faster than it improved recall: the additional passages beyond the top few were largely irrelevant, introducing noise rather than additional gold evidence.

BM25 achieved the best lexical retrieval result at F-score 0.1053, outperforming TF-IDF top-5 by approximately 0.017. The improvement reflects BM25's term frequency saturation and document length normalisation, which reduce the influence of passages that accumulate high term-frequency scores simply by being longer. However, the margin over TF-IDF was modest, and the two-stage TF-IDF top-100 plus BM25 reranking pipeline performed worse than either method alone (F = 0.0704). One explanation is that BM25 reranking applied within a small candidate set relies on local IDF statistics that differ substantially from the full-corpus statistics, and may over-rank lexically similar but semantically irrelevant passages.

Overall, the lexical retrieval results confirm that surface word matching alone is insufficient for climate fact-checking evidence retrieval. The relatively low F-scores across all lexical methods suggest that many gold evidence passages are expressed using different wording from the claim, motivating the use of neural re-ranking.

4.3 Effect of Cross-Encoder Re-ranking

Adding Cross-Encoder re-ranking on top of TF-IDF candidate retrieval produced a large improvement in evidence F-score. The single-retriever plus Cross-Encoder system achieved F = 0.2037, compared with F = 0.0883 for TF-IDF alone at the same top-5 selection threshold. This improvement demonstrates that joint encoding of claim-evidence pairs allows the model to identify semantically relevant passages that share few surface words with the claim, going beyond what sparse lexical methods can capture (Nogueira and Cho, 2019).

The dual TF-IDF retriever further improved retrieval F-score from 0.2037 to 0.2080 by providing a more diverse candidate pool. Using two complementary TF-IDF views — one with unigram features and stopword removal, and one with unigram-bigram features without stopword removal — increased the candidate pool to up to 400 unique passages per claim. This broader candidate set gave the Cross-Encoder more opportunities to surface relevant evidence that a single retriever might have missed, particularly for claims where important meaning is conveyed through short phrase patterns rather than individual content words.

4.4 Effect of Score Fusion and Adaptive Evidence Selection

Table 2 shows the top results from the hyperparameter search over score fusion weight α and evidence selection strategy. All top-ranked configurations used gap-based adaptive selection rather than fixed top-k, and all used $\alpha > 0$.

The absence of $\alpha = 0.00$ from the top 20 results indicates that score fusion consistently helped: adding a small lexical signal improved final ranking over using the Cross-Encoder score alone. The best α was 0.03, which weights the lexical score at 3% of the final score. This result supports the motivation described in Section 2.4: when multiple candidates receive similar Cross-Encoder scores near the selection boundary, a candidate that also has strong lexical overlap with the claim is preferred, providing a stable tiebreaker signal.

Config	F
gap, α=0.03, δ=0.03, min=1, max=6	**0.2438**
gap, α=0.04, δ=0.03, min=1, max=6	0.2411
gap, α=0.04, δ=0.03, min=1, max=5	0.2410
gap, α=0.06, δ=0.05, min=1, max=4	0.2386
gap, α=0.00, best	$<$0.234

Table 2: Top evidence selection configurations from the development search. All top-20 results used gap-based adaptive selection; no fixed top-k or α=0.00 configuration appeared in the top 20.

The dominance of gap-based adaptive selection over fixed top-k in the top results suggests that different claims genuinely require different numbers of evidence passages. A fixed top-k rule either selects too few passages for claims with multiple relevant pieces of evidence, or includes too many noisy passages for claims that can be verified by a single strong passage. The gap-based strategy addresses this by selecting all passages whose scores are close to the top-ranked passage, subject to minimum and maximum bounds.

4.5 Claim Classification Results

Table 3 compares two classifier training conditions evaluated on the development set. Both classifiers use the same DeBERTa architecture and are evaluated using retrieved evidence at inference time, matching the actual test-time setting where gold evidence is unavailable. The only difference is the evidence source used *during training*: one classifier is trained on gold-annotated evidence passages, while the other is trained on passages selected by our retrieval pipeline.

Evidence source	Dev accuracy
Majority label (no evidence)	0.4416
DeBERTa + gold evidence	0.4814
DeBERTa + retrieved evidence	**0.5649**

Table 3: Claim classification accuracy on the development set under different evidence conditions.

The DeBERTa classifier trained and evaluated with retrieved evidence achieved 0.5649, substantially higher than the majority-label baseline (0.4416) and also higher than the gold-evidence setting (0.4814). The fact that the retrieved-evidence classifier outperformed the gold-evidence classifier is somewhat counterintuitive: gold evidence is by definition the correct supporting or refuting passage, so one might expect it to provide a stronger classification signal. One possible explanation is that training with retrieved evidence better matches the inference distribution. The classifier trained on gold evidence encounters the exact annotated passages during training but noisier retrieved passages at inference time, creating a train-inference mismatch. Training directly on retrieved evidence may therefore teach the classifier to be more robust to the imperfect retrieval context it will face at test time.

Combining the best retrieval configuration with the DeBERTa classifier yielded a final harmonic mean of 0.3352, compared with 0.3096 when applying majority-label classification to the same retrieved evidence. This improvement confirms that replacing the majority-label placeholder with a trained classifier provides a meaningful gain in the combined metric.

4.6 Error Analysis and Limitations

Despite the improvements from neural re-ranking, the evidence retrieval F-score of 0.2383 remains relatively low, suggesting that a substantial fraction of gold evidence passages are not recovered by the system. The primary bottleneck is first-stage candidate recall: if the correct evidence is not included in the 400-passage candidate pool generated by the dual TF-IDF retrievers, the Cross-Encoder has no opportunity to rank it highly. Climate fact-checking involves claims about scientific processes and statistics that are often expressed using different terminology from the evidence, which lexical retrieval cannot bridge.

A second source of error is the non-determinism introduced during Cross-Encoder fine-tuning. Negative evidence passages are sampled from a Python `set`, whose iteration order is not guaranteed to be stable across different runtime sessions. This means that even with a fixed random seed, the negative training examples may differ between runs, producing slightly different Cross-Encoder weights and evidence rankings. This limits the exact reproducibility of the retrieval F-score across independent training runs.

A third source of error is that randomly sampled negatives are typically easy to distinguish from positive evidence, which may not adequately train the model to discriminate between hard negatives: passages that are topically related to the claim but do not actually support or refute it. This could lead to the re-ranker assigning high scores to superficially relevant passages.

For classification, the most likely source of error is confusion between labels that share similar evidence patterns. Table 4 shows the predicted and gold label distributions on the development set. The system over-predicts SUPPORTS (+25) while severely under-predicting NOT_ENOUGH_INFO (-18) and DISPUTED (-11), whose combined deficit nearly matches the SUPPORTS surplus. This suggests that claims with absent or conflicting evidence are frequently absorbed into SUPPORTS predictions. REFUTES is only marginally inflated (+4), indicating the model distinguishes negative evidence more reliably than ambiguous evidence.

Label	**Gold**	**Predicted**	**Diff**
SUPPORTS	68	93	$+25$
REFUTES	27	31	$+4$
NOT_ENOUGH_INFO	41	23	-18
DISPUTED	18	7	-11
Total	154	154	—

Table 4: Gold vs. predicted label distribution on the development set.

5 Conclusion

We proposed a multi-stage climate claim verification pipeline combining dual TF-IDF candidate retrieval, Cross-Encoder re-ranking, score fusion, adaptive evidence selection, and DeBERTa-based claim classification. Two improvements drove the bulk of performance gains. Cross-Encoder re-ranking produced the largest improvement in evidence retrieval, raising evidence F-score from 0.1053 to 0.2037. Training the classifier on retrieved rather than gold evidence produced the largest improvement in classification accuracy, raising it from 0.4814 to 0.5649. Score fusion and adaptive evidence selection provided further retrieval gains. The final system achieved an evidence F-score of 0.2383, classification accuracy of 0.5649, and harmonic mean of 0.3352 on the development set.

The main limitation is the first-stage TF-IDF retrieval: if the correct evidence is not selected into the 400-candidate pool, no later stage can recover it. This creates a hard ceiling on retrieval performance regardless of how well the Cross-Encoder re-ranks the candidates. Future work could replace TF-IDF with dense retrieval to improve candidate coverage, use hard negative mining to better train the Cross-Encoder, or jointly train retrieval and classification to reduce error propagation (Karpukhin et al., 2020; Reimers and Gurevych, 2019).

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